BSIS Program Review

Introduction

This essay is intended as a reflection of the major courses I have taken at the University of South Florida (USF) ahead of my impending graduation. In May, 2026, I will graduate with my Bachelor of Science in Information Science (BSIS) with a concentration on Data Science & Analytics (and a minor in Creative Writing.) My Information Science courses have covered concepts like programming skills, statistical analysis skills, communication skills, project management skills, and miscellaneous technical skills. Overall, this degree has given me a variety of basic knowledge and experiences in the highly interdisciplinary field of Information Science.

List of LIS Courses I've Taken at USF

While pursuing my Information Science degree at USF, I attended 13 courses dedicated to my major:

- LIS 3261 – Introduction to Information Science
- LIS 3353 – IT Concepts for Information Professionals
- LIS 4273 – Advanced Statistics & Analytics
- LIS 4800 – Introduction to Data Science
- LIS 2780 – Database Concepts
- LIS 4317 – Visual Analytics
- LIS 4761 – Introduction to Data & Text Mining
- LIS 4204 – Information Behaviors
- LIS 4414 – Information Policy & Ethics

- LIS 3361 – WWW Page Design & Management
- LIS 4370 – R Programming
- LIS 4805 – Predictive Analytics
- LIS 4934 – BSIS Senior Capstone

Information Science Core Courses Reflection

The core Information Science courses I took at USF consisted of: Intro to Information Science, IT Concepts for Information Professionals, Information Behaviors, Information Policy & Ethics, and a Senior Capstone course. Intro to Information Science gave me a general overview of the interdisciplinary field, as well as how it differs overall from subfields like Data Science or Library Science. My IT Concepts for Information Professionals course gave me ample opportunities to research the applications of Information Science across different industries, from cybersecurity to artificial intelligence. In my Information Behaviors course, we learned about the systems and challenges associated with accessing information in different domains, from public resources to private institutions. In my Information Policy and Ethics course, we discussed various ethical systems as well as their problems and implications, and how those ethical systems can be applied to real-world challenges faced by information professionals today. In my Senior Capstone course, I was instructed to create a portfolio covering an overview of everything I had learned related to my major throughout my degree.

Data Science & Analytics Concentration Courses Reflection

The Information Science courses I took dedicated to my concentration on Data Science & Analytics consisted of Intro to Data Science, Advanced Statistics & Analytics, Intro to Visual Analytics, R Programming, Intro to Data & Text Mining, and Predictive Analytics. In my Intro to

Data Science course, we were given a general overview of the field as well as a foundation in R programming for data analytics. In my Advanced Statistics & Analytics course, I learned to apply advanced statistical concepts to data using R programming, such as when different types of statistical analyses are appropriate and in what use cases. In my Intro to Visual Analytics course, I learned about basic principles of design and visually representing data in appropriate ways, including when certain types of visualizations are appropriate for certain types of data. In my R Programming course, I built upon my existing R knowledge by learning advanced mathematical and logistical programming concepts and ultimately built my own custom R package. In my Intro to Data & Text Mining course, I applied the programming and statistical skills I learned to derive meaningful insights from text-based data of various types. Finally, in my Predictive Analytics course, I got to build my own machine learning models to predictively analyze data for future trends.

Two Most Useful Assignments from the Program

Two of the most beneficial projects I was assigned to complete during my Information Science degree program were:

- Statistical Analysis Blog for my Advanced Statistics & Analytics Course:

  For this project, I was required to create a blog where I uploaded the R code I wrote for the weekly assignments. Each week, in addition to completing the steps of the statistical analysis tasks, I wrote out explanations of the statistical concepts and English translations of the written code. This helped me understand and remember important concepts that I would go on to build upon later as my knowledge of data science increased. It has also proven useful for my future career as an information professional by practicing communicating complex ideas in ways that are simple to understand.

- Video Game Industry Historical Sales Analysis for my Data & Text Mining Course:

  For this project, I got hands-on experience with text mining and data visualization techniques using R programming, as well as presenting and collaborative experience as the project team leader. I set up the cloud file sharing system, important documents, and a calendar of deadlines for all team members. I also hosted weekly meetings to check progress and put together and delivered half of the final presentation. This was useful toward my future career as an information professional by providing me with a leadership opportunity for a data-focused project.

Personal Reflection

As a matter of personal opinion, while I appreciate the exposure to the many different concepts under the umbrella of Information Science, I do not feel that this degree has adequately prepared me to enter the workforce. The courses I took were rudimentary and lacked technical depth for a technology focused STEM program. I felt that I was completing busy work that did not challenge me intellectually or teach employable skills the majority of the time. This also took significant time away from applying for internships to gain valuable real-world experience. Several of the required courses especially took considerable time without considerable gain and could have realistically been condensed into one, such as Intro to Information Science, IT Concepts, and Information Behaviors. I also felt that R Programming was an entirely redundant course, because by the time I took it, I had already been using R for over a year and a half in my other courses. This course should have been a prerequisite to others if it is going to be included as a graduation requirement and could easily have been combined with Intro to Data Science.

Further, after 600+ unsuccessful job applications, I can attest that employers for entry level data science and analysis jobs specifically ask for technologies like analyzing data with

Python, advanced SQL, Excel, Tableau, Power BI, working with big data, cloud technology, and more, which were not covered at all during this degree program. It is also extremely important to note that, despite being dedicated to data, data security, storage, and optimization practices were never discussed during this program. I am left with the impression that this program is entirely out of touch with the real-world demands of the workforce and instead tries extensively to coalesce into a basic degree that does not have a clear purpose. As an upcoming graduate applying for jobs, I am now in an impassable position that makes me both overqualified for unskilled jobs and underqualified for entry level data related roles. Practically speaking, I am even less employable now than I was before I got my bachelor's degree. Therefore, this degree seems to have prepared me for more education, but not to enter the workforce as I had originally intended.

Conclusion

Overall, this degree has given me a variety of basic knowledge and experiences in the highly interdisciplinary field of Information Science. A combination of course assignments, projects, research, and exposure to a wide range of concepts has prepared me to pursue additional education. More skills, real-world technical experience, and in some circumstances a Master's Degree will be required to feasibly pursue employment as an information professional in the current market.